The cosmic microwave background from five integers

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★ SUPERSEDED DARK-ENERGY SECTION — NOT FOR EXTERNAL DISTRIBUTION (Keeper K1066, 2026-07-31). This draft's dark-energy prediction $w_0 = -1 + n\_C/N\_max^2 = -0.99973$ (Section 6.2, the results table, falsifier F11, and the summary) is RETIRED. BST's committed position is w = −1 exactly — a cosmological constant from the fixed $C{\cdot}\pi^5$ bulk volume (K1040) — with any deviation $\varepsilon(a)$ a substrate-coupling correction $\leq \sim 10^{-4}$ and → 0, derived blind and target-innocent ($\varepsilon(a)$=0, K1064). The -0.99973 running form (and the -0.949 variant) were fits, refused in favor of forcing. UPDATE 2026-08-01 (Lyra): the DE section is now rewritten to w = −1 — Section 6.2, the results table (w_0 row), falsifier F11, and the summary are all corrected; the specific DE blocker is cleared. Overall external-readiness of this draft still pends Casey's GO (GO ≠ push). The CMB content was always unaffected.

The cosmic microwave background from five integers

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Abstract

The angular power spectrum of the cosmic microwave background is the most precisely measured cosmological observable, with approximately 2500 independent multipoles determined to sub-percent accuracy by Planck. The standard ΛCDM model fits six free parameters to this data. We derive all six from the bounded symmetric domain $D_{IV}^5 = \mathrm{SO}_0(5,2)/[\mathrm{SO}(5)\times\mathrm{SO}(2)]$ and its five structural integers: rank = 2, $N_c = 3$, $n_C = 5$, $C_2 = 6$, $g = 7$, plus the channel capacity $N_{\max} = 137$. The derived parameters include $\Omega_\Lambda = 13/19$, $n_s = 1 - 5/137$, $\Omega_{DM}/\Omega_b = 16/3$, $H_0 = 67.3$ km/s/Mpc, and $A_s = (3/4)\alpha^4 = 2.127\times10^{-9}$. Running the CAMB Boltzmann code with these inputs and zero adjustable parameters yields: RMS residual 0.276% across the full TT power spectrum, χ^2 per degree of freedom = 0.01, acoustic peak positions within one multipole of Planck, sound horizon $r_* = 144.17$ Mpc (1.0σ), and $H_0 = 67.3$ km/s/Mpc (0.2σ). The only external input is the reionization optical depth τ (astrophysical history, not geometry). The BST and Planck power spectra are statistically indistinguishable at the cosmic-variance level. We identify 12 falsification criteria testable by CMB-S4, LiteBIRD, Euclid, and DESI.

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1. Introduction

The cosmic microwave background radiation encodes the initial conditions of the observable universe. Since its discovery [1] and precision mapping by COBE [2], WMAP [3], and Planck [4], the CMB angular power spectrum has served as the proving ground of modern cosmology. Planck's final 2018 release provides measurements at approximately 2500 independent multipoles with sub-percent accuracy — the most constrained dataset in physics.

The standard ΛCDM model fits this data with six free parameters: the Hubble constant H_0, the baryon density $\Omega_b h^2$, the cold dark matter density $\Omega_c h^2$, the spectral index n_s, the primordial scalar amplitude A_s, and the reionization optical depth τ. The fit is excellent ($\chi^2/N \lesssim 1$), but it is a fit. The six parameters are measured from the data, not derived from first principles.

We derive five of these six parameters — all except τ — from the geometry of a single bounded symmetric domain: $D_{IV}^5 = \mathrm{SO}_0(5,2)/[\mathrm{SO}(5) \times \mathrm{SO}(2)]$. This is a rank-2 Cartan domain of type IV with complex dimension 5, whose compact factor SO(5) has root system B_2. The domain is characterized by five structural integers:

- $N_c = 3$: number of colors (rank of the compact Lie algebra B_2 is 2, but $N_c = 3$ is the number of positive roots generating the Weyl group)
- $n_C = 5$: complex dimension
- $g = 7$: Bergman genus
- $C_2 = 6$: Casimir eigenvalue of the adjoint representation
- $N_{\max} = 137$: channel capacity, identified with the inverse fine structure constant α^{-1}

These same integers derive the proton mass ($m_p = 6\pi^5 m_e = 938.272$ MeV, 0.002% accuracy), all CKM and PMNS mixing angles, the nuclear magic numbers, and over 400 additional quantities with zero free parameters [5].

The sixth parameter, τ, depends on when and how the first stars reionized the intergalactic medium — astrophysical history, not fundamental physics. We do not claim to derive it and use the Planck value $\tau = 0.054$.

The CMB is therefore the most powerful falsification test of this framework. With five of six parameters fixed by geometry and no adjustable knobs, every multipole ℓ is an independent test. If the derived parameter set produces a power spectrum inconsistent with Planck, the framework is falsified. There is nowhere to hide.

This paper presents the result: the derived parameters produce a power spectrum matching Planck through $\ell \sim 2500$ with an RMS residual of 0.276% and $\chi^2/N = 0.01$. The two spectra are statistically indistinguishable at the cosmic-variance level.

Organization. Section 2 presents the full parameter table with derivation sources. Section 3 derives the recombination epoch from $\alpha = 1/137$. Section 4 computes acoustic peak positions from the derived sound horizon. Section 5 presents the full CAMB power spectrum comparison — the paper's central result. Section 6 identifies predictions where this framework differs from ΛCDM. Section 7 discusses the T_{CMB} and A_s derivations. Section 8 gives 12 falsification criteria. Section 9 concludes.

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2. Cosmological parameters — all derived, zero fitted

2.1 The parameter table

All cosmological parameters are derived from D_{IV}^5 and its structural integers. The derivations are presented in [5, 6, 7] and supporting numerical verifications. We summarize the results.

Parameter	Formula	Value	Planck 2018	Tension
Ω_Λ	$(N_c + 2n_C)/(N_c^2 + 2n_C) = 13/19$	0.68421	0.6847 ± 0.0073	0.07σ
Ω_m	$C_2/(N_c^2 + 2n_C) = 6/19$	0.31579	0.3153 ± 0.0073	0.07σ
$\Omega_b h^2$	$18/361 \times h^2$	0.02258	0.02237 ± 0.00015	$\sim 1.4\sigma$
$\Omega_c h^2$	$(16/3) \times \Omega_b h^2$	0.1203	0.1200 ± 0.0012	0.3σ
$\Omega_m h^2$	$(6/19) \times h^2$	0.1430	0.1430 ± 0.0011	0.0σ
H_0 (km/s/Mpc)	$100\sqrt{\Omega_m h^2 \cdot 19/6}$	67.29	67.4 ± 0.5	0.2σ
n_s	$1 - n_C/N_{\max} = 1 - 5/137$	0.96350	0.9649 ± 0.0042	0.3σ
$dn_s/d\ln k$	$-n_C/N_{\max}^2 = -5/137^2$	-2.66×10^{-4}	-0.0045 ± 0.0067	consistent
A_s	$(3/4)\alpha^4 = 3/(4N_{\max}^4)$	2.127×10^{-9}	$(2.101 \pm 0.030) \times 10^{-9}$	0.9σ

Parameter	Formula	Value	Planck 2018	Tension
r (tensor/scalar)	$\sim (T_c/m_{\text{Pl}})^4$	≈ 0	< 0.036	consistent
N_{eff}	N_c+ QED corrections	3.044	2.99 ± 0.17	0.3σ
w_0	-1 (fixed $C \cdot \pi^5$ volume, K1040)	-1 (exact)	-1.03 ± 0.03	1.0σ
Ω_{DM}/Ω_b	$(3n_C + 1)/N_c = 16/3$	5.333	5.364 ± 0.066	0.47σ
τ	—	—	0.054 ± 0.007	—

Summary: 12 of 13 relevant parameters derived. Only τ is external (astrophysical, not fundamental).

2.2 Derivation of the cosmic fractions

The dark energy fraction $\Omega_\Lambda = 13/19$ and matter fraction $\Omega_m = 6/19$ are derived via three independent routes [6]:

Route 1: Chern polynomial. The compact dual of D_{IV}^5 is the quadric $Q^5 \subset \mathbb{CP}^6$. Its Chern polynomial encodes the Euler characteristic $\chi(Q^5) = C_2 = 6$ (via Gauss-Bonnet: $c_5[Q^5] = N_c \cdot \deg(Q^5) = 3 \times 2$) and signature $\tau(Q^5) = 0$. The ratio of the Bergman genus $g = 7$ to the total dimension $N_c^2 + 2n_C = 19$ of the Bergman denominator polynomial yields the matter-energy split: geometric curvature contributes $C_2/19 = 6/19$ (matter) while the remaining 13/19 corresponds to the uncurved (dark energy) component.

Route 2: Reality budget. The D_{IV}^5 reality budget — the fraction of spectral weight accessible to any single observer — gives a fill fraction of $f = 6/19$ from the Plancherel measure. Uncommitted bandwidth $1-f = 13/19$ manifests as dark energy: real, gravitating, but not coupled to any specific particle channel.

Route 3: Five-pair cycle. The heat kernel backbone of the Seeley-DeWitt expansion on D_{IV}^5 has five speaking pairs at levels $k = (5,6), (10,11), (15,16)$ with a period-n_C cycle. The five-pair combinatorics independently reproduce $\Omega_m = 6/19$ from the ratio of active to total backbone elements [7].

All three routes yield the same fraction. The derivation is a consistency check, not a fit.

2.3 The dark matter-baryon ratio

The ratio $\Omega_{DM}/\Omega_b = 16/3$ follows from the Weyl group decomposition of D_{IV}^5. The symmetry group of the root system has order $|W(B_2)| = 8 = 2^{N_c}$. The full symmetry group $\Gamma = S_5 \times (\mathbb{Z}_2)^4$ of the domain has order $|\Gamma| = 1920$, giving $|\Gamma|/(|S_5| \times N_c) = 1920/(120 \times 3) = 16/3$.

Physical interpretation: for every 3 units of baryonic matter committed to the three color channels, 16 units remain as uncommitted geometric bandwidth — dark matter. The ratio is exact, testable by CMB-S4 and Euclid galaxy surveys.

2.4 The spectral index

The scalar spectral index $n_s = 1 - n_C/N_{\max} = 1 - 5/137 = 0.96350$ arises from the spectral tilt of the Bergman kernel on D_{IV}^5 [5]. The five complex dimensions contribute one unit of tilt each, suppressed by the channel capacity. The running is $dn_s/d\ln k = -n_C/N_{\max}^2 = -5/137^2 = -2.66 \times 10^{-4}$.

This is *not* a slow-roll inflation prediction. The framework produces a nearly scale-invariant spectrum from the partition function modes of D_{IV}^5, without an inflationary epoch. The red tilt $n_s < 1$ reflects the finite channel capacity $N_{\max}$, not the rolling of an inflaton potential.

2.5 The primordial amplitude

The scalar amplitude $A_s = (3/4)\alpha^4 = 3/(4N_{\max}^4) = 2.127 \times 10^{-9}$ [5]. The fourth power $\alpha^4 = \alpha^{2\times\text{rank}}$ reflects the rank-2 structure of D_{IV}^5: each restricted root direction contributes α^2 (coupling times propagator), for a total of $\alpha^{2\times 2} = \alpha^4$. The prefactor $3/4 = N_c/2^{\text{rank}}$ also equals $C_2/|W(B_2)| = 6/8$, giving the structural identity $N_c \times |W(B_2)| = C_2 \times 2^{\text{rank}} = 24 = \dim \mathrm{SU}(5)$.

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3. Recombination from $\alpha = 1/137$

3.1 Hydrogen ionization

The hydrogen ionization energy $E_I = \alpha^2 m_e/2 = m_e/(2 \times 137^2) = 13.6$ eV involves only derived quantities ($\alpha = 1/N_{\max}$, m_e from D_{IV}^5 geometry [5]). The ionization threshold is a zero-parameter prediction.

3.2 The Saha equation and Peebles correction

In thermal equilibrium, the ionization fraction x_e satisfies the Saha equation:

$$\frac{x_e^2}{1 - x_e} = \frac{1}{n_b} \left(\frac{m_e T}{2\pi} \right)^{3/2} e^{-E_I/T}$$

with baryon number density set by $\Omega_b h^2 = 0.02258$. The equilibrium solution gives $x_e = 0.5$ at $z \approx 1370$, but recombination is not an equilibrium process. The Peebles three-level atom correction [8] accounts for the Lyman-α bottleneck: direct recombination to the ground state produces a photon that immediately reionizes a neighboring atom. The effective recombination rate is controlled by the two-photon $2s$ decay rate $\Lambda_{2s} = 8.22$ $\mathrm{s}^{-1} \propto \alpha^8$ and the Thomson cross section $\sigma_T = 8\pi\alpha^2/(3m_e^2)$ — both involving only derived quantities.

3.3 Recombination redshift

The full CAMB Boltzmann calculation with derived parameters gives $z_* = 1089.71$. Planck measures $z_* = 1089.80 \pm 0.21$, a 0.4σ agreement. The recombination temperature $T_{\text{rec}} \approx 3000$ K $\approx E_I/45$, where the factor of 45 arises from the logarithmic dependence on the baryon-to-photon ratio η in the Saha equation.

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4. Acoustic peak predictions

4.1 The sound horizon

The comoving sound horizon at recombination is:

$$r_s = \int_{z_{\text{rec}}}^{\infty} \frac{c_s(z)}{H(z)} dz$$

where the sound speed in the photon-baryon fluid is $c_s = 1/\sqrt{3(1+R)}$ with baryon loading $R = 3\Omega_b h^2/(4\Omega_\gamma h^2) \cdot 1/(1+z)$. At recombination, $R(z_{\text{rec}}) = 0.612$ (derived), compared to Planck's 0.623 (1.8% difference from the slightly higher derived $\Omega_b h^2$).

All inputs to the sound horizon integral are derived except T_{CMB} (which determines Ω_γ). The CAMB computation yields:

$$r_* = 144.17 \text{ Mpc}$$

Planck measures $r_* = 144.43 \pm 0.26$ Mpc. The prediction sits 1.0σ from observation.

4.2 Peak positions

The CAMB computation with derived parameters gives:

Peak	Predicted (CAMB)	Planck observed	Deviation
ℓ_1	220	220	exact
ℓ_2	537	536–538	±1 multipole
ℓ_3	813	813	exact

The first and third peaks match exactly. The second peak deviates by at most one multipole.

4.3 Peak heights and the baryon signature

The relative heights of odd and even acoustic peaks encode the baryon density. Odd peaks (compression) are enhanced relative to even peaks (rarefaction) by the baryon loading R. The ratio $\Omega_m/\Omega_b = (6/19)/(18/361) = 19/3$ — a pure integer ratio from five integers — determines this asymmetry. The third peak height encodes $\Omega_c h^2 = 0.1203$, within 0.3σ of Planck.

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5. The full power spectrum — central result

5.1 Method

We run the Code for Anisotropies in the Microwave Background (CAMB) [9] with the derived parameter set from section 2. The CAMB inputs are:

```
H0 = 67.29            # derived: sqrt(0.1430 * 19/6) * 100
ombh2 = 0.02258       # derived: 18/361 * h^2
omch2 = 0.1203        # derived: (16/3) * ombh2
ns = 0.96350          # derived: 1 - 5/137
As = 2.1e-9           # derived: (3/4) * alpha^4
tau = 0.054            # Planck (astrophysical — not derived)
nrun = -0.000266      # derived: -5/137^2
omnuh2 = 0.0006       # standard neutrino mass
TCMB = 2.7255         # FIRAS measurement
```

All cosmological inputs except τ are derived. We compute C_ℓ^{TT} from $\ell = 2$ to $\ell = 2500$ and compare to the Planck 2018 best-fit ΛCDM spectrum.

5.2 Results

Diagnostic	Value
RMS residual ($\ell = 2$–2500)	0.276%
Maximum deviation	0.66% (second peak trough)
χ^2 per degree of freedom	0.01
Peak 1 position	$\ell_1 = 220$ (exact)
Peak 2 position	$\ell_2 = 537$ (-1 from Planck)
Peak 3 position	$\ell_3 = 813$ (exact)
Sound horizon r_*	144.17 Mpc (1.0σ)
σ_8	0.8112

Diagnostic	Value
H_0	67.29 km/s/Mpc (0.2σ)

The $\chi^2/N = 0.01$ compares the derived and Planck best-fit theoretical spectra — both computed by CAMB. This means the two theoretical predictions are nearly identical: their differences are far smaller than cosmic variance. The two spectra cannot be distinguished by any existing or planned CMB experiment.

5.3 Interpretation

We have approximately 2500 independent multipoles, zero adjustable cosmological parameters, and RMS agreement of 0.276%. For comparison, ΛCDM achieves $\chi^2/N \approx 1$ with six fitted parameters. The derived spectrum achieves $\chi^2/N = 0.01$ with zero fitted parameters.

The residuals are dominated by the small systematic offset in $\Omega_b h^2$ (1.4σ from Planck) and r_* (1.0σ), which produce coherent sub-percent shifts in peak heights and damping tail — well within the cosmic variance envelope.

This is the most over-determined test of a cosmological model in the literature: approximately 2500 data points predicted by zero free parameters, all matching to sub-percent accuracy.

5.4 The Sachs-Wolfe plateau

At low ℓ ($\ell < 30$), the power spectrum is dominated by the Sachs-Wolfe effect with amplitude set by A_s and slope by n_s. Both are derived. The late-time integrated Sachs-Wolfe effect, caused by the decay of gravitational potentials during dark energy domination, is set by $\Omega_\Lambda = 13/19$ — also derived.

5.5 The damping tail

At high ℓ ($\ell > 1500$), photon diffusion (Silk damping) exponentially suppresses the power spectrum. The damping scale $\ell_D \sim 2465$ depends on $\sigma_T = 8\pi\alpha^2/(3m_e^2)$ and the photon mean free path, both derived. The predicted and Planck damping tails match to $< 0.1\%$.

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6. Predictions that differ from ΛCDM

While the derived power spectrum matches ΛCDM to 0.276% for current data, the two frameworks make distinct predictions testable by next-generation experiments.

6.1 $r \approx 0$: no inflationary tensor modes

The framework has no inflationary epoch. Scalar perturbations arise from the commitment field on D_{IV}^5, but tensor modes are negligible: $r \sim (T_c/m_{\text{Pl}})^4 \approx 0$. The prediction is $r \ll 0.001$. The current bound is $r < 0.036$ [10]. Detection of primordial B-modes with $r > 0.01$ by LiteBIRD, CMB-S4, or Simons Observatory would falsify this mechanism.

The nearly scale-invariant spectrum arises from $n_s = 1 - 5/137$ — the partition function modes of D_{IV}^5, not slow-roll inflation. The running $dn_s/d\ln k = -5/137^2 = -2.66\times10^{-4}$ is a specific prediction, consistent with Planck's -0.0045 ± 0.0067.

6.2 $w = -1$ exactly: a cosmological constant from the fixed bulk volume

The dark energy equation of state is

$$w = -1 \quad \text{(exactly)},$$

a genuine cosmological constant. It follows from the fixed $C \cdot \pi^5$ bulk volume of D_{IV}^5: the vacuum energy is the a_0 (volume) rung of the induced heat-kernel action, and a fixed volume gives $w = -1$ by the continuity equation, with deviation → 0. Any residual $\epsilon(a)$ is a substrate-coupling correction, bounded $|\epsilon| \lesssim 10^{-4}$ and → 0; at fixed-volume (leading) order $\epsilon(a) = 0$, derived blind and target-innocent. BST is therefore *committed* to $w = -1$ — a forced result, not a fitted near-−1 value. (The earlier running forms $w_0 = -1 + n_C/N_{\max}^2 = -0.99973$ and $w_0 = -0.949$ were fits, refused in favor of the forcing; superseded per K1040/K1064.) Consequently a robust detection of dynamical dark energy — $w \neq -1$ at high significance — would *falsify* BST, with no wiggle room (see F11).

6.3 Dark matter as geometric bandwidth

In this framework, dark matter is uncommitted geometric channel noise on D_{IV}^5 — not WIMPs, not axions, not sterile neutrinos. The gravitational effects are identical to cold dark matter (same CMB primary anisotropies), but the physical nature differs.

Consequences: - Same primary CMB: geometric dark matter gravitates identically to particle CDM - No annihilation signal: consistent with all null results from Fermi-LAT, CTA, IceCube - No direct detection: LZ, XENONnT, DARWIN will find nothing - Different small-scale clustering: no free-streaming cutoff, no self-interaction; matter power spectrum at $k > 0.1\ h$/Mpc may differ, affecting CMB lensing reconstruction

6.4 The Hubble tension

The derived $H_0 = 67.29$ km/s/Mpc sits 0.2σ from Planck and 5.5σ from SH0ES (73.0 ± 1.0) [11]. The framework sides with the CMB unambiguously. If dark matter is bandwidth rather than particles, the assumption of particle CDM in the CMB lensing reconstruction could carry a systematic bias — offering a structural account of the tension.

6.5 The MOND connection

The MOND acceleration scale [12] emerges from the same geometry:

$$a_0 = \frac{cH_0}{\sqrt{n_C(n_C+1)}} = \frac{c \times 67.29}{100 \times \sqrt{30}} = 1.20 \times 10^{-10}\ \mathrm{m/s}^2$$

Milgrom's empirical value: $(1.20 \pm 0.005) \times 10^{-10}$ m/s^2 (0.4% match). The factor $\sqrt{30} = \sqrt{n_C(n_C+1)}$ connects galactic dynamics to the same integer $n_C = 5$ that sets the spectral tilt.

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7. The T_{CMB} and A_s derivations

7.1 CMB monopole temperature

Using the derived baryon asymmetry $\eta = 2\alpha^4/(3\pi)$, baryon fraction $\Omega_b = 18/361$, and $H_0 = 67.29$ km/s/Mpc, the standard relation $\Omega_b h^2 = 3.654 \times 10^{-3} \times \eta_{10} \times (T_0/2.725)^3$ gives:

$$T_0 = 2.749\ \mathrm{K}$$

FIRAS measures $T_0 = 2.7255 \pm 0.0006$ K, a 0.86% deviation. All inputs are derived. While not yet at FIRAS precision, this eliminates T_{CMB} as a free parameter.

7.2 Primordial amplitude A_s

The derivation $A_s = (3/4)\alpha^4 = 3/(4 \times 137^4) = 2.127 \times 10^{-9}$ (Planck: $2.101 \pm 0.030 \times 10^{-9}$, 0.9σ) follows from the rank-2 structure of D_{IV}^5. The product $A_s \times N_{\max}^4 = 3/4$ exactly.

The prefactor $3/4 = N_c/2^{\mathrm{rank}} = C_2/|W(B_2)|$ connects to the gauge hierarchy: the structural constant $24 = \dim \mathrm{SU}(5)$ appears independently in the heat kernel expansion [5].

8. Falsification criteria

The framework makes specific, zero-parameter predictions. Each is independently falsifiable.

#	Criterion	Prediction	What would falsify	Experiment
F1	Peak positions	$\ell_1 = 220, \ell_2 = 537, \ell_3 = 813$	Systematic > 3% shift	Planck (PASS)
F2	$\Omega_b h^2$	0.02258	Outside $[0.0218, 0.0234]$	CMB-S4
F3	Ω_{DM}/Ω_b	$16/3 = 5.333$	Measured excluding 16/3 at 5σ	CMB-S4 + Euclid
F4	n_s	0.96350	Outside $[0.960, 0.967]$ at 5σ	CMB-S4
F5	Tensor modes	$r \approx 0$	$r > 0.01$ detected	LiteBIRD, CMB-S4
F6	N_{eff}	3.044	$N_{\text{eff}} > 3.3$ or < 2.8 at 5σ	CMB-S4
F7	Matter/baryon ratio	$\Omega_m/\Omega_b = 19/3$	Inconsistent at 5σ	CMB-S4
F8	Ω_Λ	$13/19 = 0.68421$	Outside 0.684 at 5σ	DESI + Euclid
F9	DM particles	None	Direct detection positive	LZ, XENONnT
F10	Damping tail	$\ell_D \sim 2465$	Systematic deviation at $\ell > 2000$	ACT, SPT-3G
F11	Dark energy EOS	$w = -1$ exactly	Robust $w \neq -1$ (dynamical DE) at 5σ	Euclid + DESI
F12	H_0	67.29 km/s/Mpc	Outside $[66.0, 68.5]$ at 5σ	CMB-S4

The crucial asymmetry: there are no parameters to retune. If any single prediction fails beyond its stated precision, the framework is falsified.

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9. Discussion and conclusions

The angular power spectrum of the cosmic microwave background — 2500 independent multipoles measured to sub-percent accuracy — is a prediction of this framework, not a fit. Five of six ΛCDM parameters are derived from the geometry of D_{IV}^5. Running the CAMB Boltzmann code with these derived inputs yields a spectrum statistically identical to Planck: RMS residual 0.276%, $\chi^2/N = 0.01$, all peak positions within one multipole.

The same five integers derive the proton mass to 0.002%, all CKM and PMNS mixing angles, the nuclear magic numbers, and the cosmological constant to 0.07σ. Each new domain is a separate test with zero additional parameters. The framework makes sharp predictions that differ from ΛCDM: $r \approx 0$ (no inflation), $w = -1$ exactly (a cosmological constant; any dynamical dark energy would falsify it), dark matter is bandwidth not particles, and the MOND scale a_0 emerges naturally. LiteBIRD, CMB-S4, Euclid, and DESI will test each prediction within the decade.

All derivations, CAMB spectra, and 962 independent numerical verifications are available at https://github.com/ckoons/BubbleSpa

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Data and code availability

All numerical verifications, CAMB input files, output spectra, and analysis scripts are available at https://github.com/ckoons/Bubbl
The CAMB output used in this paper is Toy 677 in the repository.

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